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CSA0609- Design and Analysis of Algorithms

Out of Boundary Paths

AIM

To write a Python program to find the number of paths by which a ball can move outside a grid boundary within a given maximum number of moves.

ALGORITHM

1. Start.
2. Read the grid size, maximum moves, and starting position.
3. Initialize a memoization dictionary.
4. Check the four possible directions for each move.
5. If the ball moves outside the grid, return 1.
6. If no moves remain, return 0.
7. Store and reuse previously calculated results.
8. Display the total number of paths.
9. Stop.

PSEUDOCODE

Algorithm OutOfBoundaryPaths(m, n, N, r, c)

Begin

If position is outside grid, return 1

If N = 0, return 0

If result is already stored, return result

Calculate paths in four directions

Store the result

Return result

End

PYTHON CODE

main.py

Visualize

Run

```python
def paths(m, n, N, r, c, dp):
    if r < 0 or r >= m or c < 0 or c >= n:
        return 1
    if N == 0:
        return 0

    if (r, c, N) in dp:
        return dp[(r, c, N)]

    dp[(r, c, N)] = (
        paths(m, n, N-1, r-1, c, dp) +
        paths(m, n, N-1, r+1, c, dp) +
        paths(m, n, N-1, r, c-1, dp) +
        paths(m, n, N-1, r, c+1, dp)
    ) % (10**9 + 7)

    return dp[(r, c, N)]


m = int(input("Rows: "))
n = int(input("Columns: "))
N = int(input("Maximum moves: "))
r = int(input("Starting row: "))
c = int(input("Starting column: "))

print("Number of paths:", paths(m, n, N, r, c, {}))
```

OUTPUT

Output

```
Rows: 2
Columns: 2
Maximum moves: 2
Starting row: 0
Starting column: 0
Number of paths: 6
```

COMPLEXITY ANALYSIS

Let m be the number of rows, n be the number of columns, and N be the maximum number of moves.

Time Complexity: $O(m \times n \times N)$

Space Complexity: $O(m \times n \times N)$

RESULT

Thus, the Python program successfully calculates the number of paths by which a ball can move outside the grid boundary using Dynamic Programming with Memoization.